

DDA3020 Homework 1 Solution

Due date: March 09, 2025

1 Written Problems (50 points)

1.1. (Closed-Form Solution, 25 points) Given a ridge regression with data $\mathbf{X} \in \mathbb{R}^{n \times m}$, $\mathbf{y} \in \mathbb{R}^n$, where n is the number of data, m is the number of attributes used for prediction, under the assumption that data $\mathbf{X}$ is centered, i.e., $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = \mathbf{0}$, show that the closed-form solution of the cost function

$$J(\mathbf{w}, w_0) = (\mathbf{y} - \mathbf{X}\mathbf{w} - w_0\mathbf{1})^T(\mathbf{y} - \mathbf{X}\mathbf{w} - w_0\mathbf{1}) + \lambda\mathbf{w}^T\mathbf{w}$$

is

$$\hat{w}_0 = \bar{y}$$

$$\mathbf{w} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

Solution:

$$\begin{aligned} J(\mathbf{w}, w_0) &= (\mathbf{y} - \mathbf{X}\mathbf{w} - w_0\mathbf{1})^T(\mathbf{y} - \mathbf{X}\mathbf{w} - w_0\mathbf{1}) + \lambda\mathbf{w}^T\mathbf{w} \\ &= \mathbf{y}^T\mathbf{y} + \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w} - 2\mathbf{y}^T\mathbf{X}\mathbf{w} + \lambda\mathbf{w}^T\mathbf{w} - 2w_0\mathbf{1}^T\mathbf{y} + 2w_0\mathbf{1}^T\mathbf{X}\mathbf{w} + w_0\mathbf{1}^T\mathbf{1}w_0 \end{aligned}$$

Note that given $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = \mathbf{0}$ last three terms should be

$$\begin{aligned} w_0\mathbf{1}^T\mathbf{y} &= w_0n\bar{y} \\ w_0\mathbf{1}^T\mathbf{X}\mathbf{w} &= w_0 \sum_i \mathbf{x}_i^T \mathbf{w} = n\bar{\mathbf{x}}^T \mathbf{w} = 0 \\ w_0\mathbf{1}^T\mathbf{1}w_0 &= nw_0^2 \end{aligned}$$

Calculate gradient w.r.t. w_0 , we have

$$\frac{\partial}{\partial w_0} J(\mathbf{w}, w_0) = -2n\bar{y} + 2nw_0 = 0 \Rightarrow \hat{w}_0 = \bar{y}$$

Calculate gradient w.r.t $\mathbf{w}$, we have

$$\frac{\partial}{\partial \mathbf{w}} J(\mathbf{w}, w_0) = 2\mathbf{X}^T\mathbf{X}\mathbf{w} - 2\mathbf{X}^T\mathbf{y} + 2\lambda\mathbf{w} = 0 \Rightarrow \mathbf{w} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

1.2. (Support Vector Machine, 25 points) Consider a dataset with 2 points in 1D: $(x_1 = 0, y_1 = -1)$ and $(x_2 = 1, y_2 = 1)$. Consider mapping each point to 3D using the feature vector $\phi(x) = [1, \sqrt{2}x, x^2]^\top$. (This is equivalent to using a second-order polynomial kernel.) The max margin classifier has the form:

$$\begin{aligned} \min \|\mathbf{w}\|^2 \quad \text{s.t.} \\ y_1 (\mathbf{w}^\top \phi(x_1) + w_0) \geq 1 \\ y_2 (\mathbf{w}^\top \phi(x_2) + w_0) \geq 1 \end{aligned}$$

- (i) Write down a vector that is parallel to the optimal vector $\mathbf{w}$. (5 points)
- (ii) What is the value of the margin achieved by this $\mathbf{w}$? Hint: recall that the margin is the distance from each support vector to the decision boundary. (5 points)
- (iii) Solve for $\mathbf{w}$, using the fact that the margin is equal to $1/\|\mathbf{w}\|$. (5 points)
- (iv) Solve for w_0 using your value for $\mathbf{w}$ and the optimization problem above. Hint: the points will be on the decision boundary, so the inequalities will be tight, think about the geometry of 2 points in space, with a line separating one from the other. (5 points)
- (v) Write down the discriminant function $f(x) = w_0 + \mathbf{w}^\top \phi(x)$ as an explicit function of x . (5 points)

Solution:

- (i) We are given the following feature vectors:

$$\phi(x_1) = [1, 0, 0]^T, \quad \phi(x_2) = [1, \sqrt{2}, 1]^T.$$

A vector parallel to the optimal weight vector $\mathbf{w}$ can be:

$$\phi(x_2) - \phi(x_1) = [0, \sqrt{2}, 1]^T.$$

- (ii) The margin is half of the distance between $\phi(x_1)$ and $\phi(x_2)$:

$$d = \sqrt{(1-1)^2 + (\sqrt{2}-0)^2 + (1-0)^2} = \sqrt{0+2+1} = \sqrt{3}.$$

Therefore, the margin is:

$$\text{margin} = \frac{1}{2} \times d = \frac{\sqrt{3}}{2}.$$

- (iii) From (i), we know that a vector parallel to the optimal weight vector $\mathbf{w}$ is:

$$\phi(x_2) - \phi(x_1) = [0, \sqrt{2}, 1]^T.$$

Therefore, we can assume that the optimal weight vector $\mathbf{w}$ has the form:

$$\mathbf{w} = [0, \sqrt{2}a, a]^T,$$

where a is a scalar to be determined, we have:

$$\frac{\sqrt{3}}{2} = \frac{1}{\|\mathbf{w}\|}, \quad \|\mathbf{w}\| = \frac{2}{\sqrt{3}}.$$

$$\|\mathbf{w}\| = \sqrt{0^2 + (\sqrt{2}a)^2 + a^2} = \sqrt{2a^2 + a^2} = \sqrt{3a^2} = \sqrt{3} \cdot |a|.$$

From $\|\mathbf{w}\| = \frac{2}{\sqrt{3}}$, we have:

$$\sqrt{3} \cdot |a| = \frac{2}{\sqrt{3}}, \quad |a| = \frac{2}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} = \frac{2}{3}.$$

Thus, $a = \pm \frac{2}{3}$.

To determine the correct sign of a , we use the constraints from the optimization problem:

$$y_1(\mathbf{w}^T \boldsymbol{\phi}(x_1) + w_0) \geq 1,$$

$$y_2(\mathbf{w}^T \boldsymbol{\phi}(x_2) + w_0) \geq 1.$$

Substitute $\mathbf{w} = [0, \sqrt{2}a, a]^T$ and the feature vectors:

$$\boldsymbol{\phi}(x_1) = [1, 0, 0]^T, \quad \boldsymbol{\phi}(x_2) = [1, \sqrt{2}, 1]^T.$$

For x_1 (with $y_1 = -1$):

$$-(\mathbf{w}^T \boldsymbol{\phi}(x_1) + w_0) \geq 1 \quad \Rightarrow \quad -w_0 \geq 1 \quad \Rightarrow \quad w_0 \leq -1.$$

For x_2 (with $y_2 = 1$):

$$\mathbf{w}^T \boldsymbol{\phi}(x_2) + w_0 \geq 1 \quad \Rightarrow \quad \sqrt{2}a \cdot \sqrt{2} + a \cdot 1 + w_0 \geq 1.$$

Simplify:

$$2a + a + w_0 \geq 1 \quad \Rightarrow \quad 3a + w_0 \geq 1.$$

Given $w_0 \leq -1$, a must be positive to satisfy the inequality. Thus:

$$a = \frac{2}{3}.$$

The optimal weight vector is:

$$\mathbf{w} = [0, \sqrt{2} \cdot \frac{2}{3}, \frac{2}{3}]^T = [0, \frac{2\sqrt{2}}{3}, \frac{2}{3}]^T.$$

(iv) The inequalities in (iii) for w_0 are:

1. Constraint for x_1 with $y_1 = -1$:

$$-w_0 \geq 1 \quad \Rightarrow \quad w_0 \leq -1.$$

2. Constraint for x_2 with $y_2 = 1$:

$$3a + w_0 \geq 1 \quad \Rightarrow \quad 3 \cdot \frac{2}{3} + w_0 \geq 1 \quad \Rightarrow \quad 2 + w_0 \geq 1 \quad \Rightarrow \quad w_0 \geq -1.$$

3. Combine Constraints:

$$w_0 \leq -1 \quad \text{and} \quad w_0 \geq -1.$$

Therefore, the only possible value for w_0 is:

$$w_0 = -1.$$

(v) Finally, the discriminant function is:

$$f(x) = \mathbf{w}^T \phi(x) + w_0 = \frac{2}{3}x^2 + \frac{4}{3}x - 1.$$

2 Programming (50 points)

2.1. (Logistic Regression, 25 points) In this question, we will implement multi-class logistic regression using gradient descent. Follow code template provided in *logistic_assignment1.ipynb* to solve this question. Noticably, this question should not be solved by using *sklearn* package, aiming to promote your understanding on gradient descent. A mathematical derivation of gradient is required in this problem. Provide your solution in your written answers.

Solution: Code implementation can be referred to *logistic_assignment1_solution.ipynb*. Detailed derivation of gradient in multi-class logistic regression is listed below.

Given data $\{\mathbf{x}_i, y_i\}_1^N$, $\mathbf{x}_i \in \mathbb{R}^{M \times 1}$, $y_i \in \mathbb{R}^{1 \times C}$. The logistic regression first calculate logit $\mathbf{z}_i = \mathbf{x}_i^T \mathbf{w} \in \mathbb{R}^{1 \times C}$, where $\mathbf{w} \in \mathbb{R}^{M \times C}$ is the parameter we would like to calculate gradient on. Based on $\mathbf{z}_i$, softmax function is used to get the probability that this sample is predicted as some class. $\hat{y}_{ic} = \frac{\exp(\mathbf{z}_{ic})}{\sum_{j=1}^C \exp(\mathbf{z}_{ij})}$. The cost function is $\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log \hat{y}_{ic}$, where we convert ground-truth label y into one-hot representation. Focus on one data $(\mathbf{x}, y)$, corresponding logit $\mathbf{z}$ and predicted probability $\hat{y}$, where subscript i is omitted.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \hat{y}} &= -\frac{1}{n} \sum_{c=1}^C y_c \cdot \frac{1}{\hat{y}_c} \\ \frac{\partial y_c}{\partial \mathbf{z}_j} &= \begin{cases} \hat{y}_c - \hat{y}_c \hat{y}_j & \text{if } c = j \\ -\hat{y}_c \hat{y}_j & \text{otherwise} \end{cases} \\ \frac{\partial \mathbf{z}}{\partial \mathbf{w}} &= \mathbf{x} \end{aligned}$$

Consider that $y_i = 1$ if and only if i is the ground truth label and $y_i = 0$ otherwise.

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \frac{1}{n} (\hat{y} - y) \mathbf{x}$$

Since we have many data and gradient contribution of every data is independent, we can use this one-data solution to construct the gradient on all data.

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \frac{1}{n} \mathbf{X}^T (\hat{\mathbf{Y}} - \mathbf{Y})$$

2.2. (Support Vector Machine, 25 points) In this question, we will explore the use of Support Vector Machines (SVM) for both linear and non-linear tasks using the *sklearn* library, as outlined in the *SVM_assignment1.ipynb* file. By following the instructions in the notebook, we will implement both types of SVMs to gain a foundational understanding of how to apply SVMs.

Solution: Code implementation can be referred to *SVM_assignment1_solution.ipynb*.